Yash Sharma

Backend & AI Engineer

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Experience

AI/ML Engineer Intern, *Tridevum Technology* 02/2026 – 08/2026
Delhi-NCR, India

- Backend & API Integration: Built async FastAPI microservices integrating OpenAI and Gemini APIs, with rate limiting and error logging.
- RAG & Vector Search: Engineered RAG pipelines using ChromaDB and semantic chunking for context-aware retrieval over custom datasets.
- Prompt Engineering & UI Prototyping: Optimized prompts using JSON Schema outputs to minimize model hallucinations; developed interactive demo dashboards in Streamlit.

Education

Bachelor of Technology (B.Tech) in Computer Science & Engineering, 07/2023 – Present
R.D. Engineering College Ghaziabad
Expected Aug 2027 Cumulative GPA: 8.5 / 10.0

Senior Secondary (Class XII) – Science Stream, *Paras Public School* 07/2022 – 04/2023
Aggregate Score: 73.8% Noida

Secondary (Class X), *C.C.S. Mem. Public School* 07/2020 – 04/2021
Aggregate Score: 73.8% Ghaziabad

Skills

Core Development
Python, Java, DSA, SQL, OOP, RESTful Microservices, AsyncIO

Testing & Deployment
Unit Testing, Docker, CI/CD Basics, Git/Github, Postman

Data Storage
PostgreSQL, MySQL, Vector DBs (ChromaDB, FAISS)

DevOps & Tooling
Version Control, Linux Administration, VS Code

AI/GenAI
FastAPI, LangChain, LangGraph, RAG Pipelines

Languages

- English
- Hindi

Projects

PrepMind AI: AI-Powered JEE Preparation Platform
Python, FastAPI, LangGraph, PostgreSQL, pgvector, React, TypeScript

- RAG-Based Chatbot: Built a LangGraph tutor (query rewriting, pgvector retrieval, grounding check) over 129 NCERT/JEE PDFs, with page-level citations and live answer streaming via Server-Sent Events.

- Daily Practice with Streaks: Developed a personalised daily set of 20 questions (14 weak-topic and 6 revision) using per-chapter weakness tracking and a streak system
- Exam Simulator: Built an adaptive JEE Main mock-exam engine that assembles a 75-question, 180-minute paper, adjusting difficulty across 3 levels from the student's past accuracy and generating questions on demand when the bank runs short.

HCare — Offline-First Clinical Documentation & On-Device AI Platform
TypeScript, React, Vite, Web Workers, Whisper, IndexedDB, Python, SQL, Docker
- Multimodal Clinical Capture: Built an asynchronous Web Worker inference pipeline (Whisper speech-to-text, image preprocessing, OCR, local LLM, and vector embeddings) to transcribe medical voice notes and digitize lab reports entirely offline.
- Hybrid Case-File Extraction: Developed a structured extraction engine combining deterministic rules, medical lexicons, and local LLM generation to populate patient records with field-level provenance and a human-in-the-loop review queue.
- Encrypted Offline Sync: Engineered a client-side cryptographic vault and encrypted blob store over IndexedDB with background synchronization, diff/merge conflict resolution, role-based access control, and audit logging.

Achievement

Code in Python, Think in AI (Vol. 1)
Co-Author
- Co-authored with Kripa Shankar Sharma; wrote core modules bridging Python fundamentals (OOP, async design) with practical AI mental models and algorithm design.
- Developed hands-on code examples and structured problem sets guiding readers from basic scripting to production-grade coding practices.

Code in Python, Think in AI (Vol. 2)
Upcoming Book (In Progress)
- Authoring a comprehensive guide on end-to-end enterprise ML lifecycles, covering data quality, feature pipelines, experiment tracking, and model monitoring.
- Designing production-grade capstone projects showcasing model serving via FastAPI, containerization using Docker, and modern bridges to Generative AI, RAG, and Agentic systems.